The Cloud Architecture Handbook
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Page 1: Architecture Overview
READ_V2 is a serverless reader designed for zero base monthly cost.
Files are delivered directly from Cloud Storage to browser viewports.
FastAPI serves strictly as the control plane for metadata and auth.

Page 2: Progress Synchronization Strategy

In PDF files, pages are fixed geometry. Progress is stored as an integer (’42’). The client reader debounces progress updates to every 15 seconds. Immediate flushing occurs on reader close or browser visibility change.

Page 3: Google Cloud Platform Always-Free Tier

All resources are pinned strictly to us-central1 (Iowa).
Cloud Storage: 5 GB Standard Storage.
Cloud Firestore: 50,000 document reads, 20,000 document writes per day.
Cloud Run: Scales to zero instances when idle ($0 base recurring cost).